

Module 04: Enumeration

Certified Ethical Hacker (CEH v13) — Lab Report

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Course:	CEH v13 — Module 04: Enumeration
Lab Duration:	2 hours, 23 minutes (Expected: 2 hours)
Result:	10 / 10 — Passed (100%)

This report documents the enumeration techniques performed against the isolated CEH practice lab environment (network range 10.10.1.0/24, domain CEH.com), covering NetBIOS, SNMP, LDAP, DNS, and SMTP enumeration, along with the final lab assessment results.

1. Executive Summary

This report summarizes the enumeration exercises completed as part of CEH v13 Module 04. Enumeration is the process of extracting usernames, machine names, network shares, services, and other information from a target system after initial reconnaissance. All activities in this report were carried out against a designated, isolated training lab environment for educational purposes.

The lab was completed with a final score of **10/10 (100%)** in a total duration of **2 hours and 23 minutes**, against an expected duration of 2 hours.

#	Task	Result
1	NetBIOS enumeration — identify shared folder on Windows Server	\\WINDOWS11\CEH-Tools
2	SNMP enumeration using SnmpWalk — community string option	snmpwalk -v1 -c public
3	LDAP enumeration using AD Explorer — Domain Controller IP	10.10.1.22
4	LDAP enumeration using AD Explorer — userPrincipalName for Jason@CEH.com	jason@CEH.com
5	NFS enumeration using RPCScan / SuperEnum — NFS service port	2049

2. NetBIOS Enumeration

2.1 Objective

Identify the shared folder/driver available on the Windows Server (Windows11 host, 10.10.1.11) using NetBIOS enumeration tools.

2.2 Commands and Findings

The `nbtstat -c` command was used to display the NetBIOS remote cache name table, resolving the host name **WINDOWS11** to IP address **10.10.1.11**. The `net use` command was then used to list active connections, revealing a mapped network drive (Z:) pointing to the shared resource **\\WINDOWS11\CEH-Tools** on the Microsoft Windows Network.

```
NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
          [-r] [-R] [-RR] [-s] [-S] [interval] ]

-a (adapter status) Lists the remote machine's name table given its name
-A (Adapter status) Lists the remote machine's name table given its
                      IP address.
-c (cache)           Lists NBT's cache of remote [machine] names and their IP addresses
-n (names)           Lists local NetBIOS names.
-r (resolved)        Lists names resolved by broadcast and via WINS
-R (Reload)          Purges and reloads the remote cache name table
-S (Sessions)        Lists sessions table with the destination IP addresses
-s (sessions)        Lists sessions table converting destination IP
                      addresses to computer NETBIOS names.
-RR (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

RemoteName Remote host machine name.
IP address  Dotted decimal representation of the IP address.
interval    Redisplays selected statistics, pausing interval seconds
             between each display. Press Ctrl+C to stop redisplaying
             statistics.

C:\Users\Administrator>nbtstat -c

Ethernet 2:
Node IpAddress: [10.10.1.19] Scope Id: []

NetBIOS Remote Cache Name Table

      Name                Type      Host Address      Life [sec]
-----
WINDOWS11                <20>    UNIQUE           10.10.1.11        326

C:\Users\Administrator>net use
New connections will be remembered.

Status      Local      Remote      Network
-----
OK          Z:         \\WINDOWS11\CEH-Tools  Microsoft Windows Network
The command completed successfully.
```

Figure 1: `nbtstat -c` and `net use` output showing the resolved host and shared folder **\\WINDOWS11\CEH-Tools**.

2.3 Additional NetBIOS Enumeration (AI-Assisted)

NetBIOS enumeration was additionally performed against 10.10.1.11 using ShellGPT (an AI shell assistant), which generated and executed the command `nmblookup -A 10.10.1.11 && nbtscan 10.10.1.11`. The output confirmed the host name **WINDOWS11** and workgroup membership, consistent with the manual `nbtstat` results above.

```
Enter your AI Activation Key: 08630443c5184c1d9f241b9ffb089013
ShellGPT configuration updated successfully.
Environment variables set: file
AZURE_API_BASE=https://apiservice5lh2czy2mmcta.azure-api.net/ecc-sqmp-v1/
AZURE_API_VERSION=2024-09-01-preview
Verifying environment variables...
AZURE_API_BASE: https://apiservice5lh2czy2mmcta.azure-api.net/ecc-sqmp-v1/
AZURE_API_VERSION: 2024-09-01-preview
Executing sgpt command...
Hello! How can I assist you today? ☺
[~][root@parrot]-[/home/attacker]
→ #sgpt --shell "Perform NetBIOS enumeration on target IP 10.10.1.11"
nmblookup -A 10.10.1.11 && nbtscan 10.10.1.11
[E]xecute, [D]escribe, [A]bort: E
Looking up status of 10.10.1.11
    WINDOWS11  Winda <00> -      B <ACTIVE>
    WORKGROUP   Trans <00> - <GROUP> B <ACTIVE>
CERVIS WORKGROUP <20> -      B <ACTIVE>
Hack WORKGROUP <1e> - <GROUP> B <ACTIVE>
Appl WORKGROUP <1d> -      B <ACTIVE>
```

Figure 2: AI-assisted NetBIOS enumeration via ShellGPT, confirming host WINDOWS11.

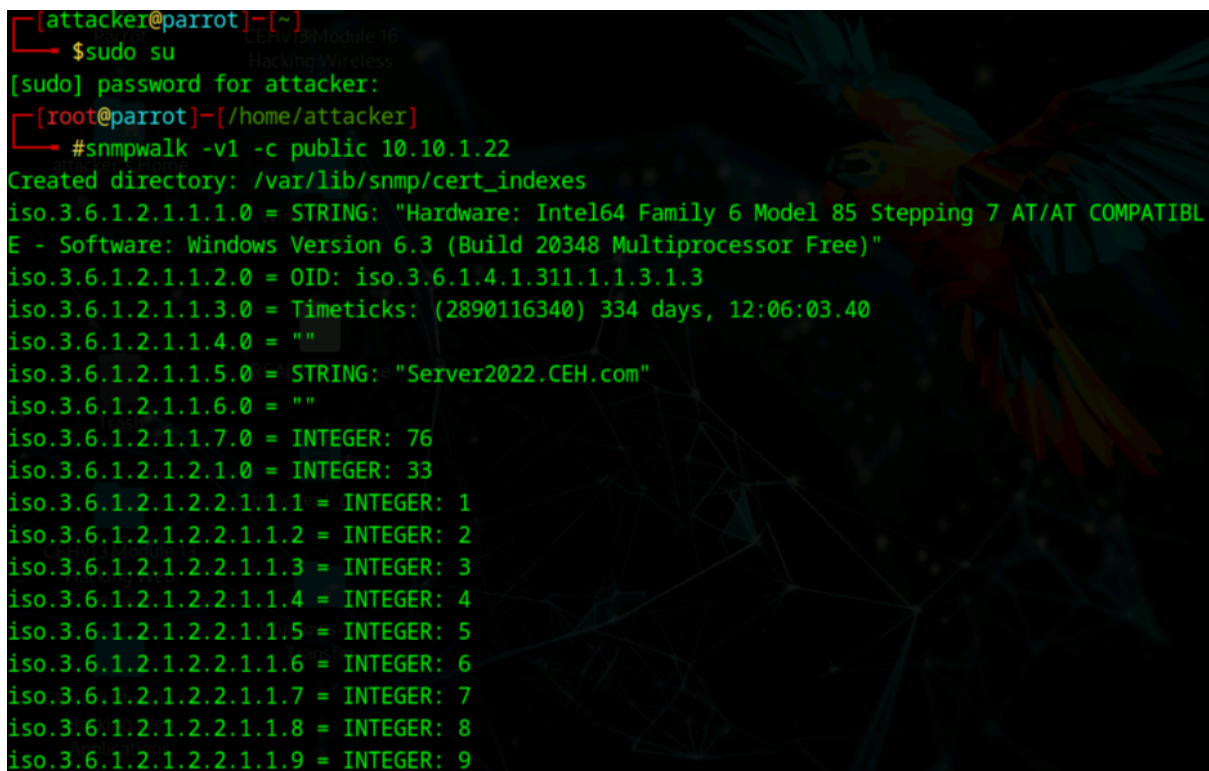
3. SNMP Enumeration

3.1 Objective

Perform SNMP enumeration on the Windows Server 2022 machine (10.10.1.22) using SnmpWalk to extract system information via the default community string.

3.2 Commands and Findings

The command `snmpwalk -v1 -c public 10.10.1.22` was executed, using SNMP version 1 with the community string **public**. The output disclosed detailed system information, including the OS (Windows Server 2022, Build 20348), the system name **Server2022.CEH.com**, system uptime, and a walk of further OIDs — demonstrating that the SNMP service was misconfigured with a default/guessable community string.

A terminal window with a dark background and a parrot illustration on the right. The terminal shows a user 'attacker' at 'parrot' running 'sudo su' to become root. Then, the command '#snmpwalk -v1 -c public 10.10.1.22' is executed. The output lists various SNMP OIDs and their values, including hardware details, software version (Windows Server 2022), system name (Server2022.CEH.com), and system uptime (334 days, 12:06:03.40).

```
[attacker@parrot]~  
$sudo su  
[sudo] password for attacker:  
[root@parrot]~/home/attacker  
#snmpwalk -v1 -c public 10.10.1.22  
Created directory: /var/lib/snmp/cert_indexes  
iso.3.6.1.2.1.1.1.0 = STRING: "Hardware: Intel64 Family 6 Model 85 Stepping 7 AT/AT COMPATIBL  
E - Software: Windows Version 6.3 (Build 20348 Multiprocessor Free)"  
iso.3.6.1.2.1.1.2.0 = OID: iso.3.6.1.4.1.311.1.1.3.1.3  
iso.3.6.1.2.1.1.3.0 = Timeticks: (2890116340) 334 days, 12:06:03.40  
iso.3.6.1.2.1.1.4.0 = ""  
iso.3.6.1.2.1.1.5.0 = STRING: "Server2022.CEH.com"  
iso.3.6.1.2.1.1.6.0 = ""  
iso.3.6.1.2.1.1.7.0 = INTEGER: 76  
iso.3.6.1.2.1.2.1.0 = INTEGER: 33  
iso.3.6.1.2.1.2.2.1.1.1 = INTEGER: 1  
iso.3.6.1.2.1.2.2.1.1.2 = INTEGER: 2  
iso.3.6.1.2.1.2.2.1.1.3 = INTEGER: 3  
iso.3.6.1.2.1.2.2.1.1.4 = INTEGER: 4  
iso.3.6.1.2.1.2.2.1.1.5 = INTEGER: 5  
iso.3.6.1.2.1.2.2.1.1.6 = INTEGER: 6  
iso.3.6.1.2.1.2.2.1.1.7 = INTEGER: 7  
iso.3.6.1.2.1.2.2.1.1.8 = INTEGER: 8  
iso.3.6.1.2.1.2.2.1.1.9 = INTEGER: 9
```

Figure 3: snmpwalk -v1 -c public output disclosing OS, hostname, and system details of 10.10.1.22.

4. LDAP Enumeration (Active Directory Explorer)

4.1 Objective

Use AD Explorer (AD Explorer) to connect to the domain controller and enumerate Active Directory objects, identifying the Domain Controller's IP address and the userPrincipalName of a specific user.

4.2 Domain Controller Identification

AD Explorer was connected to the CEH.com domain. The connection path displayed CN=Domain Controllers, CN=Users, DC=CEH, DC=com, 10.10.1.22 [Server2022.CEH.com], confirming the Domain Controller's IP address as 10.10.1.22. The directory tree further listed domain objects including user accounts, groups (Domain Admins, Enterprise Admins, Schema Admins, etc.), and service accounts under the CEH.com domain.

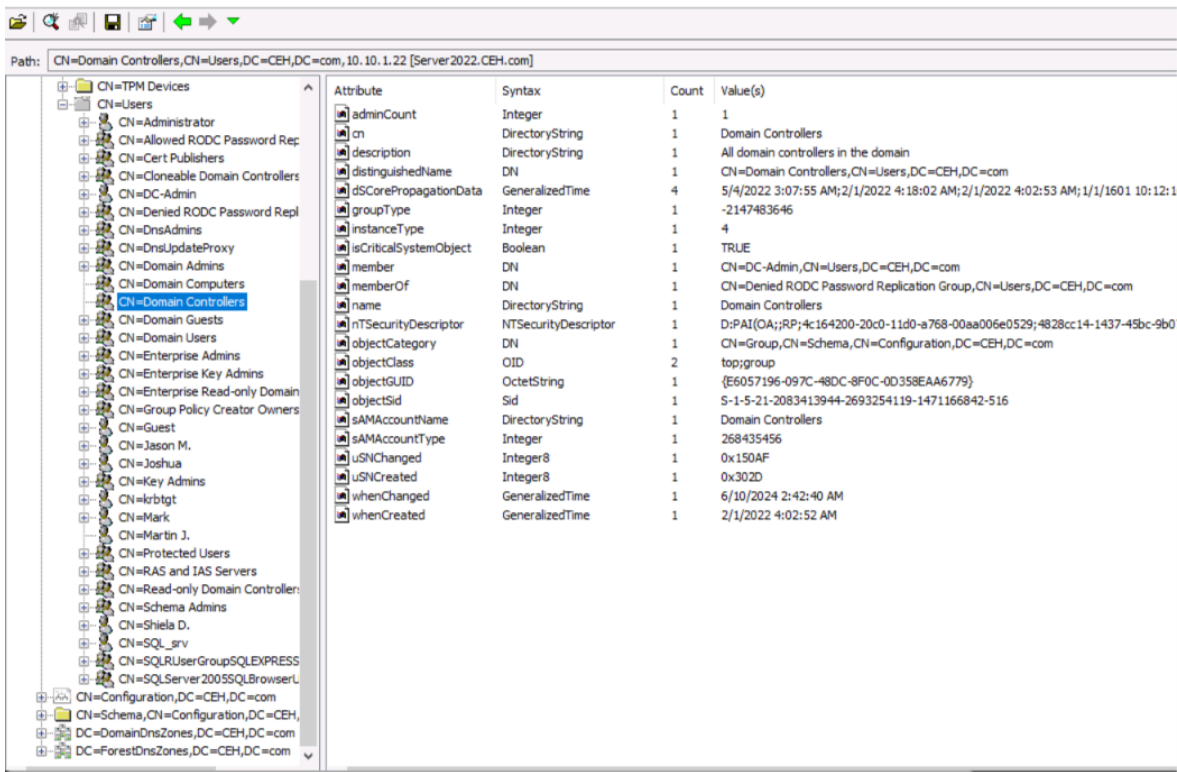


Figure 4: AD Explorer showing the Domain Controllers container and DC IP address 10.10.1.22 (Server2022.CEH.com).

4.3 User Attribute Enumeration — userPrincipalName

The Attribute Properties for the user object CN=Jason M. , CN=Users, DC=CEH, DC=com were inspected. The userPrincipalName attribute value returned was jason@CEH.com, along with other visible attributes such as sAMAccountName (jason), badPwdCount, and account control flags.

accountExpires	Integer8	1	0x7FFFFFFFFFFFFFFF
adminCount	Integer	1	1
badPasswordTime	Integer8	1	6/7/2024 3:19:35 AM
badPwdCount	Integer	1	2
cn			
codePage			
countryCode			
displayName			
distinguishedName			
dSCorePropagationD			
givenName			
initials			
instanceType			
lastLogoff			
lastLogon			
logonCount			
memberOf			
name			
nTSecurityDescriptor			
objectCategory			
objectClass			
objectGUID			
objectSid			
primaryGroupID			
pwdLastSet			
sAMAccountName	DirectoryString	1	jason
sAMAccountType	Integer	1	805306368
userAccountControl	Integer	1	66048
userPrincipalName	DirectoryString	1	jason@CEH.com
uSNCreated	Integer8	1	0x3241
uSNChanged	Integer8	1	0x3241

Attribute Properties

Attribute: userPrincipalName
Object: CN=Jason M.,CN=Users,DC=CEH,DC=com
Syntax: DirectoryString
Schema: CN=User-Principal-Name,CN=Schema,CN=Configuration,DC=CEH,DC=com Go to

Values:
jason@CEH.com

OK

Figure 5: AD Explorer Attribute Properties dialog showing userPrincipalName = jason@CEH.com for user Jason M.

5. NFS Enumeration

5.1 Objective

Perform NFS enumeration against 10.10.1.19 using RPCScan and SuperEnum to identify the port used by the NFS service.

5.2 Findings

Enumeration using RPCScan and SuperEnum against 10.10.1.19 confirmed that the Network File System (NFS) service was exposed on the standard NFS port, **2049/tcp**. This result is reflected in the final lab assessment score sheet (Figure 9).

6. Additional Reconnaissance Activities

Beyond the graded lab tasks, the following supplementary enumeration activities were performed against the lab targets to build a broader footprint of exposed services.

6.1 DNS Enumeration — Zone Transfer Attempts

A DNS zone transfer was attempted against the domain **certifiedhacker.com**, a standard target domain used in CEH lab exercises. Using `nslookup` with `set querytype=soa`, the SOA record was retrieved, identifying the primary name server **ns1.bluehost.com**. A subsequent zone listing attempt (`ls -d ns1.bluehost.com`) was refused by the DNS server. A follow-up attempt using `dig @ns1.bluehost.com www.certifiedhacker.com axfr` from the Parrot OS attacker machine likewise failed, confirming that the zone transfer was properly restricted by the name server.


```

Microsoft Windows [Version 10.0.22000.469]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>nslookup
Default Server:  dns.google
Address:  8.8.8.8

> set querytype=soa
> certifiedhacker.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
certifiedhacker.com
    primary name server = ns1.bluehost.com
    responsible mail addr = dnsadmin.box5331.bluehost.com
    serial = 2025100600
    refresh = 86400 (1 day)
    retry = 7200 (2 hours)
    expire = 3600000 (41 days 16 hours)
    default TTL = 300 (5 mins)
> ls -d ns1.bluehost.com
[dns.google]
*** Can't list domain ns1.bluehost.com: Server failed
The DNS server refused to transfer the zone ns1.bluehost.com to your computer. If this
is incorrect, check the zone transfer security settings for ns1.bluehost.com on the DNS
server at IP address 8.8.8.8.

```

Figure 6: nslookup SOA query for certifiedhacker.com and a refused zone listing attempt.

```

; OPT PSEUDOSECTION:
EDNS: version: 0, flags:; udp: 512
; QUESTION SECTION:
19: 445, Tool: nmap_smb-system-info
www.certifiedhacker.com.45, To IN nmap NS os-discovery
Trying for 10.10.1.19: 445, Tool: nmap_smb-security-mode
; ANSWER SECTION:
19: 445, Tool: nmap_smbv2-enabled
www.certifiedhacker.com. 10397 IN CNAME certifiedhacker.com.
certifiedhacker.com. 21600 IN NS ns1.bluehost.com.
certifiedhacker.com. 21600 IN NS ns2.bluehost.com.
[0] in function 'error'
; Query time: 113 msec/nmap/nse/main.lua:829: in local 'get_chosen_scripts'
; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)sin:100/1364: in main chunk
; WHEN: Sun Jul 12 15:04:09 EDT 2026
; MSG SIZE rcvd: 111
; HITTING!
-[root@parrot]~# 19: 445, Tool: nmap_smb-brute
#dig @ns1.bluehost.com www.certifiedhacker.com axfr
; global options: +cmd
Transfer failed.

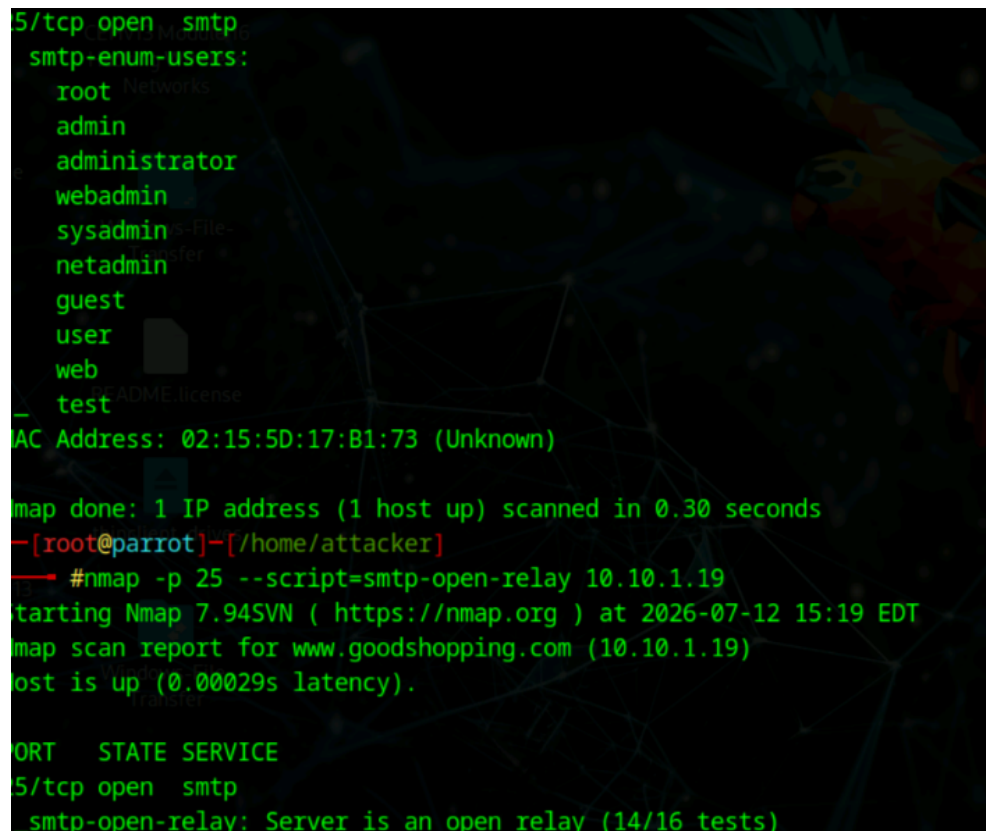
```

Figure 7: dig AXFR zone transfer attempt against ns1.bluehost.com for www.certifiedhacker.com — transfer failed.

6.2 SMTP Enumeration

SMTP enumeration was performed against 10.10.1.19 (www.goodshopping.com). An Nmap NSE scan using the smtp-enum-users script enumerated a list of common valid-looking usernames (root, admin, administrator, webadmin, sysadmin, netadmin, guest, user, web, test) on port 25/tcp. A follow-up scan

with `nmap -p 25 --script=smtp-open-relay 10.10.1.19` confirmed the SMTP server was configured as an **open relay**, passing 14 of 16 relay tests — a significant misconfiguration that could allow unauthorized mail relaying.



```
5/tcp open  smtp
smtp-enum-users:
  root
  admin
  administrator
  webadmin
  sysadmin
  netadmin
  guest
  user
  web
  test
MAC Address: 02:15:5D:17:B1:73 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
-[root@parrot]~[/home/attacker]
#nmap -p 25 --script=smtp-open-relay 10.10.1.19
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-12 15:19 EDT
Nmap scan report for www.goodshopping.com (10.10.1.19)
Host is up (0.00029s latency).

PORT      STATE SERVICE
5/tcp     open  smtp
smtp-open-relay: Server is an open relay (14/16 tests)
```

Figure 8: Nmap smtp-enum-users output and smtp-open-relay results confirming an open mail relay on 10.10.1.19.

7. Final Assessment Results

The lab platform confirmed successful completion of all Module 04: Enumeration tasks with a final score of 10/10 (100%), completed in 2 hours and 23 minutes against an expected duration of 2 hours.

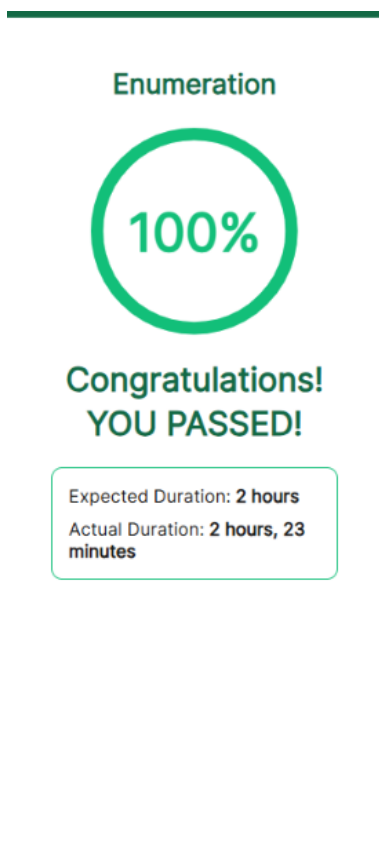


Figure 9: Lab completion summary — 100% score, 2 hours 23 minutes actual duration.

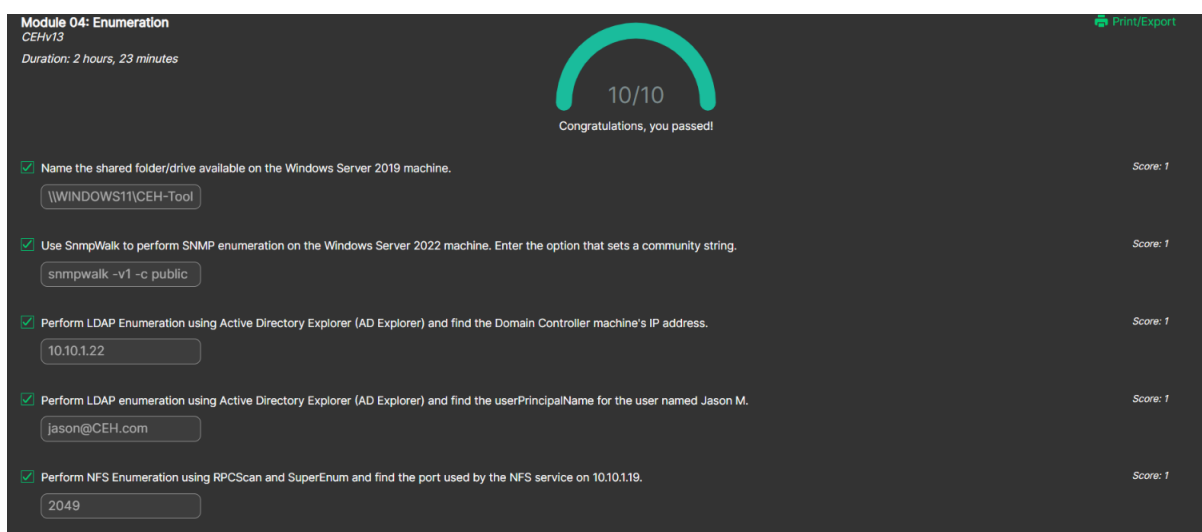


Figure 10: Module 04 Enumeration scoresheet — 10/10, listing each task and the submitted answer.

8. Conclusion

This lab demonstrated how multiple enumeration techniques — NetBIOS, SNMP, LDAP/Active Directory, NFS, DNS, and SMTP — can be combined to build a detailed profile of a target network's hosts, shares, user accounts, and misconfigured services. Key misconfigurations identified in the lab environment included a guessable SNMP community string (public), an openly accessible network share, and an SMTP server acting as an open relay. In a real-world environment, these findings would support recommendations such as disabling or restricting SNMP, applying least-privilege share permissions, enforcing NFS access controls, and disabling open mail relaying.